

Cryptography Algorithms and Verilog Implementation 130330

Making Our Own Verilog Design (Assignment 2)

Date: Sunday, March 15, 2026

Submission:	By not more than 2 people together
Deadline: Final report	Sunday, April 12, 2026 (17:15) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

1. About the Exercise:

You are already familiar with this particular design and its testbench... This time we are doing the design and the testbench in Verilog.

2. Specifications:

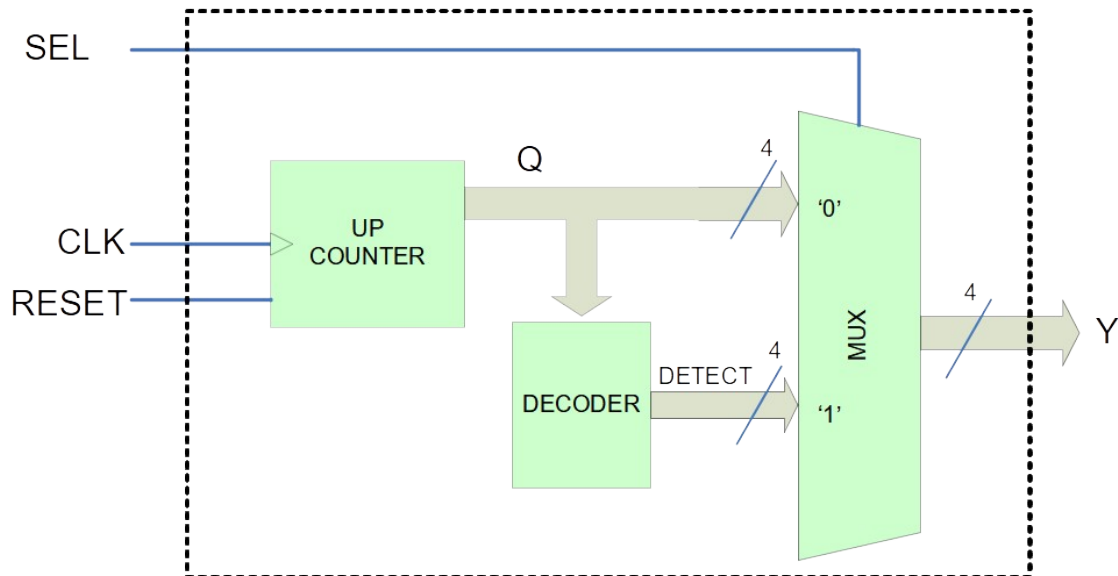
The required system is based on a counter, decoder and a mux as follows:

- The counter is a “Free running” 4-bit up counter, with synchronous, active high reset.
- The decoder receives the output of the counter, and produces an internal DETECT signal according to the following table:

Decoder's output	Behavior
DETECT(0)	1: When the decoder's input is in the range of 0 – 3 0: Otherwise
DETECT(1)	1: When the decoder's input is in the range of 4 – 7 0: Otherwise
DETECT(2)	1: When the decoder's input is in the range of 8 – 11 0: Otherwise
DETECT(3)	1: When the decoder's input is in the range of 12 – 15 0: Otherwise

- The mux, controlled by the SELECT (“SEL”) input, selects whether the design's output (Y) shall reflect the output of the counter or the output of the decoder.

The following diagram illustrates the design's functionality:



The design's I/O signals are given in the following table:

Signal	Direction	Description
CLK	IN	Master clock signal
RESET	IN	Active high synchronous reset
SEL	IN	Controls the source of the output signal as follows: 0: Y reflects the output of the counter directly 1: Y reflects the output of the internal decoder
Y[3:0]	OUT	4 bit output signal

In addition, the design contains the following internal signals:

Signal	Description
Q[3:0]	The output of the internal counter
DETECT[3:0]	The output of the internal decoder

Note: The counter does not contain a TC (Terminal Count) signal.

3. ATP:

- Look into your notes and write the ATP for this design

4. Design:

- Write the Verilog code for the design. Implement the design using Verilog “always” and “assign” statements as applicable

5. Test Bench

1. Write a Verilog testbench for the design
2. Monitor the internal Q and DETECT signals directly from the testbench, by using Verilog XMR

6. Synthesis and Place & Route

1. Prepare an XDC file relevant for the design, for a target Artix7 device on the Basys3 board.
2. Synthesize and Place & Route the design, targeting the design to the Artix7 device of the Basys3 board.

Note: For synthesis, you may use either Precision or Vivado.

7. Your Final Report Shall Include:

1. The Verilog source code of the design
2. The Verilog testbench for the design
3. Transcript with the \$monitor results
4. Waveforms including both testbench and internal signals
5. Synthesis report, RTL schematics and Technology schematics, area and timing report.
6. XDC file for your design.
7. Vivado’s device view and pin location results. Utilization and timing results.